Software Requirements Specification Template

Software Engineering

The following annotated template shall be used to complete the Software Requirements Specification (SRS) assignment.

Template Usage:
Text contained within angle brackets (‘<’, ‘>’) shall be replaced by your project-specific information and/or details. For example, <Project Name> will be replaced with either ‘Smart Home’ or ‘Sensor Network’.

Italicized text is included to briefly annotate the purpose of each section within this template. This text should not appear in the final version of your submitted SRS.

This cover page is not a part of the final template and should be removed before your SRS is submitted.

Smart Cinema System

Software Requirements Specification

Version 1.0

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Group #13
Khang Do
Minh Nguyen
Phu Pham

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Instructor: Gus Hanna, Ph.D.
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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

Signature	Printed Name	Title	Date
	<Your Name>	Software Eng.	
	Dr. Gus Hanna	Instructor, CS 250	

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1. Introduction

The purpose of this Software Requirements Specification is to define the Cinema Systems' scope, user roles, functional/non-functional requirements, interfaces, use cases and analysis models. The Cinema System provides customer access to searching for movies and their showtimes; selecting seats; purchasing tickets; and viewing their booking history. Additionally, employees have the ability to sell/verify tickets; manage movie scheduling; cancel purchases; and generate reports.

1.1 Purpose

This SRS is a definition of the cinema system requirements; it will serve as a reference for Cinema System Developers, Testers, Cinema Managers and Instructors to link system characteristics, use cases, models and tests using the requirement ID.

1.2 Scope

The Cinema System supports one cinema with multiple auditoriums. Customers can view showtimes, select seats, pay, receive QR tickets, and cancel bookings when allowed. Staff can sell and verify tickets, manage schedules, and handle sales and refunds. Version 1.0 excludes concessions, loyalty programs, streaming, and multi-cinema management.

1.3 Definitions, Acronyms, and Abbreviations

Term	Meaning
Showtime	One screening of one movie in one auditorium at a scheduled time.
Hold and booking	A hold reserves seats temporarily. A booking records seats, contact, price, and lifecycle status.
QR ticket	A ticket containing an unpredictable token encoded as a QR code.
Idempotent operation	Repeating a request or callback does not repeat its business effect.
Reconciliation	Checking provider records to resolve an uncertain payment or refund.

1.4 References

William Y. Arms, Scenarios and Use Cases, Cornell University CS 5150. Supplied as lecture-use-cases.pdf; publication date not stated. Used for actors, scenarios, and use case descriptions.

E-Store Project Software Requirements Specification, Version 4.0, Group-1, April 16, 2007. Supplied as SRS4.0.doc. Used as an example of detailed functional requirements.

Shock Force Software Team, Software Requirements Specification, October 20, 2009. Supplied as SoftwareRequirementsTemplate.docx. Used as an example of operational scenarios and analysis models.

1.5 Overview

Chapter 2 describes the product, users, assumptions, and business rules. Chapter 3 defines detailed requirements and use cases. Chapter 4 presents analysis models. Chapter 5 controls requirement changes. Appendix A maps requirements to acceptance tests; Appendix A.2 provides sample data and an open-decision register.

2. General Description

2.1 Product Perspective

The Cinema System is a browser-based application that has an inventory of seats which are available for both customer use and staff use. The Cinema System connects to external payment processing systems as well as email systems; however, the projection equipment is separate from the Cinema System.

2.2 Product Functions

The system supports customer booking and cancellation, box-office sales and payments, ticket admission, and management of movies, seats, showtimes, prices, staff roles, and operations.

2.3 User Characteristics

The system includes Customers, Cashiers, Ushers, and Managers with different access levels, plus external payment and email services for transactions and notifications.

2.4 General Constraints

This system will need an Internet connection in order to access the server for managing time, price, seat availability and ticket sales. This system does not use any particular programming language or type of database.

2.5 Assumptions and Dependencies

Version 1.0 supports one cinema using USD and local venue time. Bookings contain 1–8 seats for one showtime, with 5-minute holds and no partial cancellation or seat exchange. Sales stop at show start, while cancellations are allowed at least 30 minutes before the show if tickets are unused. Managers can cancel future shows and issue refunds. Admission is valid from 30 minutes before the show until the movie ends, with one entry per ticket. Pricing includes seat price, fees, and tax, and showtimes must include at least 20 minutes of turnaround. Card and cash refunds must be properly recorded, and payment or email failures must not create invalid bookings.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system should be simple to use. Customers can find showtimes, review orders, buy tickets, receive confirmation, and view booking history, while staff only access features needed for their roles.

3.1.2 Hardware Interfaces

The system will operate in common web browsers across all computer types, tablet devices, and mobile phone devices. Tickets are able to be scanned with the use of a device's camera, or manually entered into the system. Box-office tickets may be printed from a standard printer.

3.1.3 Software Interfaces

The system securely integrates with payment, e-mail, and data services; it also prohibits duplicate payment transactions, duplicate ticket transactions, and duplicate seat transactions.

3.1.4 Communications Interfaces

The system utilizes HTTPS to verify event providers, prevents duplicate actions from occurring, monitors the status of external services utilized by the system, and requires on-line ticket verification.

3.2 Functional Requirements

Each feature below follows the course template: introduction, inputs, processing, outputs, and error handling. Detailed processing statements carry the IDs. All externally supplied data must be validated before it changes a record.

3.2.1 Account and role management

3.2.1.1 Introduction

Customer identity and staff access are managed separately.

3.2.1.2 Inputs

Email, password, recovery token, staff account, and assigned role.

3.2.1.3 Processing

Customers will be able to browse all of the movie releases they want to see, find and sort through the available showtimes, and view important showtime information (such as time, auditorium, price, and how many seats are left) without being able to choose sold-out, canceled, unpublished, or already-started showtimes.

3.2.1.4 Outputs

An authenticated session, verified customer profile, or audited staff-role change.

3.2.1.5 Error Handling

Reject duplicate account emails and expired tokens. Return a generic response to a recovery request so it does not reveal whether an account exists.

3.2.2 Movie browsing and showtime search

3.2.2.1 Introduction

Customers find screenings that can still be booked.

3.2.2.2 Inputs

Movie title, date, language, format, and optional filters.

3.2.2.3 Processing

The system shows available seats, holds selected seats for up to five minutes, releases expired holds, and rechecks availability before purchase.

3.2.2.4 Outputs

Matching movies and showtimes with current availability.

3.2.2.5 Error Handling

Invalid dates produce a validation message. A stale listing never overrides the final inventory check.

3.3 Use Cases

3.3.1 Use case #1

Customers are able to establish, check validity of, log-in to, and modify their accounts. Managers have the ability to assign/remove employee roles with all modifications being recorded and implemented at the time of modification.

3.3.2 Use case #2

Customers are able to search through movie listings; sort by showtime; see pricing and availability; and access a seating chart for any available showing.

3.3.3 Use case #3

The Customer or Cashier selects 1–8 seats for an available showtime. The system checks all selected seats at once and either holds all of them or none. A successful hold lasts 5 minutes and includes a price snapshot and hold reference. If any seat becomes unavailable or the hold expires, the reservation fails or is released.

3.4 Classes / Objects

3.4.1 Account

Stores the account information of users (name, email etc.), authentication data (password hash), user profile information (date of birth), role assignment to each user (admin, editor), and account status.

3.4.2 Movie

Stores all details related to movies; e.g. - title, length/runtime, rating (IMDB), language, format (DVD, Bluray), archival status.

3.4.3 Auditorium and Seat

Stores auditorium layouts and seat plans for all auditoriums. It also contains seat label information for each seat within an auditorium and accessibility information.

3.4.4 Showtime

Stores the schedule of showtimes in auditoriums with associated pricing, start times, and current status of the showtime.

3.5 Non-Functional Requirements

Unless stated otherwise, the following targets are Must requirements. Performance tests use 500 concurrent browsing sessions, 50 concurrent checkout sessions, up to 10 auditoriums with 300 seats each, and a measured client round-trip latency no greater than 100 ms. Third-party delays are measured separately from internal processing.

3.5.1 Performance

The system will deliver fast and reliable performance through the quick processing of all catalog, showtime, seat-hold, admission and payment confirmation requests in just a couple of seconds; as well as making sure that any expired seats are made available for sale as soon as possible and the available seats remain accurate at all times.

3.5.2 Reliability

The system will provide for reliability and consistency of transaction processing through prevention of duplicate seating assignments/admissions; preservation of confirmed booking/payment/refund data following system failures; automatic review of uncertain payments; and timely restoration/recovery from backups while avoiding the creation of incomplete/inconsistent record sets.

3.5.3 Availability

The system will provide high availability with a focus on being available at all times. It should be able to continue functioning even if the email service is down or there is an issue with the payment service. Also, when a customer attempts to complete their booking transaction and fails, the system should save their booking data so they can attempt to book again later without seeing an incorrect "successful" message.

3.5.4 Security

The system will provide security for all user accounts, reservations, payments, and audits by providing Role-Based Access Control (RBAC) to ensure that only authorized users can view or modify specific types of data; securely storing user passwords; using protected and expiring token for authentication; automatically timing out user sessions after a period of inactivity; limiting the number of failed attempts to log in before an account is locked; authenticating all third-party event providers; and limiting guests from making reservation directly on behalf of another user.

3.5.5 Maintainability

The system will be able to be maintained through configuration of policy settings at a high level; log all reservations and transactions in such a manner as to allow for review later on; require an update to testing, reviewing and rolling back procedures after any significant changes to the requirements.

3.5.6 Portability

The System should be portable; it will also support major Web browsers. It should also have a responsive design to work well on different size screens.

Additionally, the System should have Keyboard Navigation, Clear Text Labels/ Non-Color-Only Indicators, Multiple Ticket Validation Methods and Standard Browser-Based Printing (No Proprietary Hardware).

3.6 Inverse Requirements

The system will not validate an unpaid online reservation or a credit card reservation based on success messages from the browser. The system will not sell the same seat at the same show time to two different active reservations.

3.7 Design Constraints

Sales on line and in the ticket office shall be carried out via one authoritative system for allocation and data storage supporting transactions and unique entries. Client-side validations are optional. Money amounts in th 1.không e database shall be stored in integers denoting cents, and time stamps in UTC instants. Dates displayed or reported shall be presented using the venue time zone.

3.8 Logical Database Requirements

The system must ensure all event, request, transaction, refund, and ticket identifiers are unique. Each successful payment can generate at most one refund. Tickets are retrieved using token hashes, with encrypted token backups for reprinting or delivery. Booking records must preserve the original movie, showtime, seat, payment, and billing details so later catalog changes do not alter past receipts.

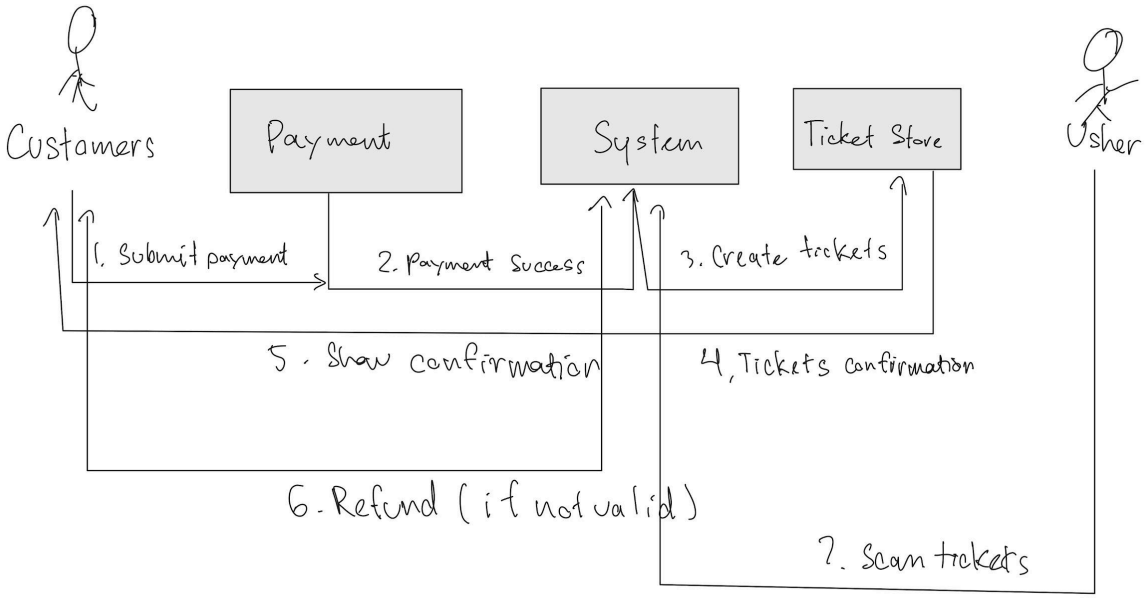
3.9 Other Requirements

The customer confirmation and cancellation screens must show the cutoff for cancellation using the time zone of the venue and the refund status in clear language. Staff support must provide explanations for the following cases expired hold, uncertain payment, double scan, printing error, and cash refund on hold.

4. Analysis Models

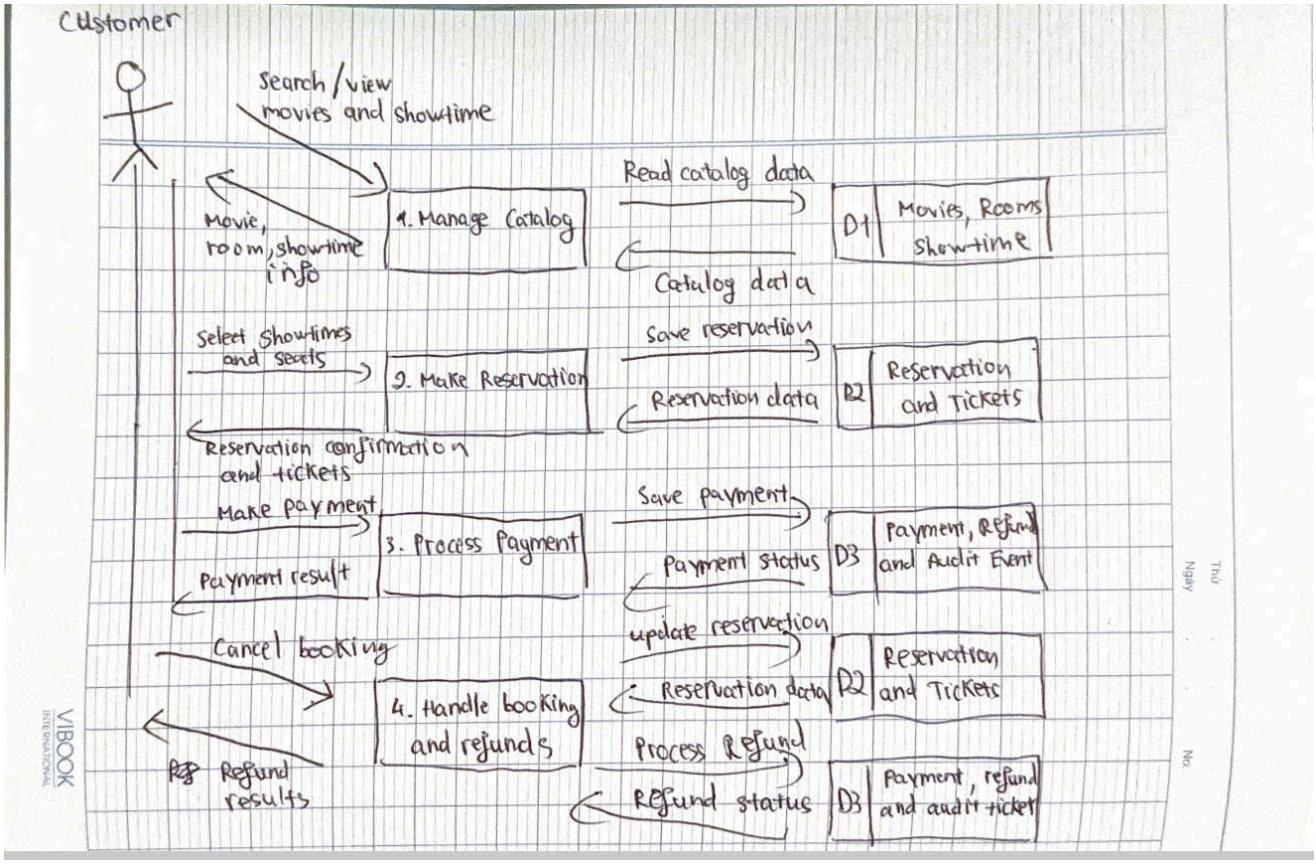
4.1 Sequence Diagrams

Payments confirm bookings only when seats and showtimes remain valid; late payments are refunded. Tickets can be admitted only once, while duplicate scans are rejected and logged.



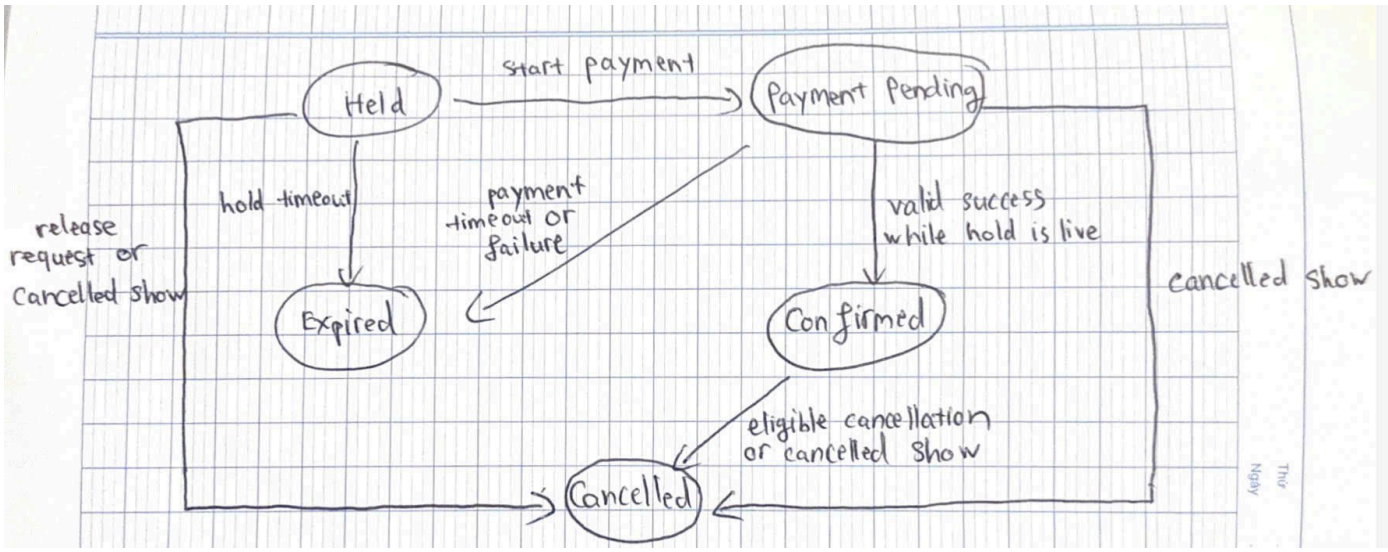
4.2 Data Flow Diagrams (DFD)

Data Flow Diagram is divided into two diagrams: customers and employees' perspectives on the four processes. The database D1 contains information about movies, rooms, and show times; D2 contains information about reservations and tickets; D3 contains information about payments, refunds, and audit information.



4.3 State-Transition Diagrams (STD)

Booking State Life Cycle starts at Seat Hold stage. From there, Booking moves on to Payment Pending if it's a card transaction or to Confirmed if it’s cash. Booking cannot go past the Expired or Cancelled stages. The Refund state is tracked via a different Refund object.



5. Change Management Process

Change requests may be submitted by a customer representative, a developer, tester or even the instructor. Change request provides for recording the change ID, reason, related requirements, suggested text changes and expected benefit. Verbal change requests must be written down prior to changing baseline. The author analyzes impact on business policies, use cases, data, diagrams, acceptance tests and scope. Migration effort related to existing holds, bookings, tickets and payment is identified.

A. Appendices

A.1 Appendix 1

With regards to testing at a fundamental level, the system must undergo tests to determine if the users are able to self-register with the system; log-in to the system; search for movies and show times; reserve seating; and cancel their reservations.

A.2 Appendix 2

This appendix has sample data for use in testing the system. Sample data includes Smart Cinema, Auditorium 1, a sample film showing at 7:00pm, a five-minute seat hold, booking from one to eight seats and QR code tickets. These are examples to test the system and will not create additional system requirements.